Substrate Carries, Architecture Pays: Check-In-Level Embeddings for Multi-Task POI Prediction

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Abstract. We introduce *Check2HGI*, a graph embedding for Point-of-Interest (POI) prediction that produces one vector per user check-in instead of one fixed vector per place, so the same café visited in the morning and in the evening feeds the model with two different inputs. In Multi-Task Learning (MTL) over Location-Based Social Networks (LBSNs), recent work reports only modest joint gains when next-category and next-region prediction share a graph embedding, suggesting that the embedding, not the loss-balancing strategy, is the bottleneck. Across five state splits of the Gowalla dataset, per-visit context lifts category macro-F1 by 14–29 percentage points at every state, while joint multi-task training improves category accuracy at the cost of region accuracy, the classic multi-task tradeoff.

Keywords: Multi-Task Learning · Point-of-Interest Prediction · Graph Embeddings.

1 Introduction

Location-Based Social Networks (LBSNs), such as Gowalla [4], record sequences of geo-tagged user *check-ins*, where each check-in stores when, where, and what kind of place a user visited. In this context, a Point of Interest (POI) is a physical place with semantic meaning, such as a restaurant, a shopping mall, an airport, or a park [12].

From these traces, two prediction tasks recur in the literature. **Next-category prediction** forecasts the semantic category of the next visit (e.g. *Food*, *Shopping*, *Travel*) and feeds content-aware recommendation and intent inference. **Next-region prediction** forecasts the next visit's spatial *region* (a coarse geographic cell that may contain many POIs, not an individual POI) and feeds demand forecasting and short-horizon mobility planning.

Because both tasks read the same mobility data, Multi-Task Learning (MTL) [2,13] is a natural fit: the standard appeal is that a shared representation lets one task's signal benefit the other and lift both. In practice the two tasks have different natures. Predicting a category is a coarse semantic decision shaped by per-visit context (intent, time of day, what the user did just before); predicting a region

is a fine-grained spatial ranking driven by short-horizon transitions and spatial regularities. Forcing one shared representation to serve both can either help or hurt, the well-known risk of *negative transfer* [13].

Two recent studies on this exact task pair report the negative outcome. The first applies hard-parameter-sharing MTL on top of a POI-stable graph embedding (one vector per place, learned with DGI [14]) and finds only marginal joint gains, with most differences within standard deviations; it diagnoses a *representation mismatch*, where one shared encoder learns a compromise specialised enough for neither head [12]. The follow-up keeps the same MTL recipe but decomposes the monolithic embedding into independent spatial, temporal and categorical sub-encoders, and recovers substantial category-side gains across three states [10]. Together they suggest the central factor is the embedding the multi-task model consumes, not the loss-balancing recipe itself.

Both prior responses kept the embedding's *emission rate* fixed (one vector per place) and changed *which* features were encoded. We test an orthogonal axis: how often the embedding emits a vector. POI-stable embeddings such as HGI [5] emit a single vector per place, so two visits to the same café at 8:00 and at 22:00 produce identical inputs. A check-in-level embedding instead emits one vector per visit, so the two visits can differ. We hypothesise that this per-visit context is the load-bearing property for the easier head (next-category), and that, once the upstream embedding (the *substrate*, by contrast with the downstream multi-task *backbone*) is fixed at the right granularity, joint training behaves like the classic MTL tradeoff: a small gain on the easier task at the cost of the harder one.

To test this hypothesis, we propose **Check2HGI**, a check-in-level extension of HGI [5], and run a controlled five-state ablation on user-disjoint splits of Gowalla, covering Alabama (AL), Arizona (AZ), Florida (FL), California (CA) and Texas (TX). We compare Check2HGI against per-place HGI under *matched-head single-task ceilings*: the same task-specific head trained alone on each substrate, so any difference between rows is attributable to the substrate rather than to head capacity. The findings are asymmetric across the two tasks. Per-visit context lifts next-category macro-F1 by +14 to +29 percentage points (pp) at every state; on next-region, per-place HGI ties or marginally exceeds Check2HGI. Joint multi-task training over Check2HGI then adds a small category lift at four of five states and pays a 7 to 17 pp cost on next-region top-10 accuracy at every state, the expected MTL tradeoff now made explicit.

The main contributions of this paper are as follows.[1]

- C0. We propose *Check2HGI*, a check-in-level contextual graph embedding that extends HGI's hierarchical contrastive objective with a fourth emission level (Check-in → POI → Region → City), so that the same place visited twice can yield two distinct vectors.
- C1. A controlled five-state matched-head ablation that separates the substrate's contribution from head capacity. Check2HGI lifts single-task cate-

[1] To support reproducibility, the full artefact set (code, fold splits, seeds, per-fold transition priors, and regression tests) is released at `https://anonymous.4open.science/r/PoiMtlNet-FE6A/`.

gory macro-F1 over per-place HGI by +14 to +29 pp at every state (paired Wilcoxon $p = 0.0312$ each, 5/5 folds positive); on next-region, the same substrate ties or marginally trails per-place HGI. A pooled-vs-canonical counterfactual replicates at Alabama and Arizona, attributing ∼72 % and ∼64 % of the category gap to per-visit context.

- C2. A direct characterisation of the multi-task tradeoff under the substrate established by C1: with the substrate fixed, joint multi-task training adds a small category lift at four of five states and pays a sign-consistent 7 to 17 pp cost on next-region top-10 accuracy at every state. Drop-in balancers and a head-side fix do not close the gap, indicating a structural cost of the cross-attention shared backbone interacting with the region head rather than a loss-recipe artefact.

Sections 2, 3, and 4 cover related work, method, and setup. Section 5 shows that the substrate is asymmetric across the two tasks: per-visit context carries next-category at every state, while per-place embeddings tie or marginally trail it on next-region. With the substrate fixed, joint multi-task training delivers a small category lift at four of five states and pays the expected region cost. Section 6 attributes the substrate gain through a per-visit counterfactual, ablates drop-in balancers and a head-side fix, and adds a methodological note on cross-attention loss-side ablations. Sections 7 and 8 close.

2 Related Work

Three lines of work bear on our study: graph-based POI embeddings that supply the substrate, multi-task learning balancers that sit on top of any joint architecture, and prior multi-task work on POI prediction tasks. We summarise each line briefly and position our contribution against it; a fourth note on cross-attention multi-task models is deferred to the methodological discussion in Section 6.

2.1 Graph-based POI embeddings

Substrate choice is the load-bearing axis our study tests. Early POI representations learn one vector per place from co-visit statistics, treating the POI as the atomic unit, and category-aware extensions inject category supervision into the same objective. Self-supervised graph contrastive methods then sharpened the representation: Deep Graph Infomax (DGI) [14] maximises mutual information between local node embeddings and a global graph summary, and Hierarchical Graph Infomax (HGI) [5] adapts the contrast across POI, region, and city graphs to capture multi-scale urban structure. DGI and HGI both emit *POI-stable* vectors: the same place visited twice yields the same representation. The substrate we use, *Check2HGI*, is a check-in-level extension of HGI's contrastive objective: contrastive views are sampled at the granularity of a single check-in, and the visit's temporal and co-visit context perturbs the view, so the same POI visited twice can yield two distinct vectors. This per-visit emission rate is the property our substrate axis tests; the recipe is described in our anonymous code release.

2.2 Multi-task learning balancers

A second body of work studies how to combine task losses or gradients once the architecture is fixed. Recent proposals include FAMO [7], which adapts loss weights from per-task progress, and Aligned-MTL [11], which aligns gradient directions through an independent-component decomposition. The dense-prediction survey [13] aggregates the line and reports that no balancer dominates across task pairs; the tradeoff identified by Caruana [2] and quantified by Maninis et al. [9] through the Δ_m joint score remains the operating regime. Our setup uses a fixed static-weight aggregation with $\lambda_{\text{cat}} = 0.75$ and ablates against FAMO and Aligned-MTL in Section 6.

2.3 Prior POI multi-task work

A third line is concept-aligned with our task pair. The human-mobility representation model [3] jointly models attribute and trajectory signals through shared encoders, and STAN [8] addresses the next-location subproblem with spatio-temporal self-attention; we adapt STAN for our region head and add a training-fold-only additive $\alpha \cdot \log T$ prior in the spirit of GETNext [15], but our head is not a faithful reproduction of any one prior model. ReHDM [6] extends the hypergraph line with a region-aware dual-level model and serves as our concept-aligned external next-region baseline. The two studies closest to the present work, [12] and [10], target the precise next-category and next-POI pairing on POI-stable graph embeddings; they are summarised in §1. Both responses kept the embedding's per-POI emission rate fixed and varied *which* features to encode; our study varies the *granularity* axis instead, by changing how often the embedding emits a vector.

3 Method

We describe the two halves of our system, followed by the training protocol that joins them. The first half is the **substrate** *Check2HGI*: a graph embedding that turns each user check-in into a contextual vector by extending HGI's place-level hierarchy with a finer level beneath POI. The second is the **backbone**: a multi-task model that consumes those vectors through two parallel input streams bridged by cross-attention before a shared residual block and per-task heads. In this work, a check-in is defined as a tuple $c = (p, r, t)$ with POI p, region r, and timestamp t; trajectories are split into non-overlapping length-nine windows predicting the category and region of the tenth check-in ($L = 9$).

3.1 The Check2HGI substrate

HGI [5] encodes urban geography at three levels (POI $\rightarrow$ region $\rightarrow$ city) and emits one vector per place, so revisits inherit the same representation. Check2HGI replaces this place-level base with a check-in-level base where each visit is a

node carrying its category and a sin/cos timestamp, and adjacent visits in a user's chronological sequence are linked under exponential temporal decay; the hierarchy becomes Check-in → POI → Region → City. A two-layer graph convolutional network (GCN) encodes the visit nodes; multi-head attention pools (*Checkin2POI*, then POI-to-region) plus a region-adjacency GCN propagate to the region level; an area-weighted sigmoid yields the city embedding.

Training maximises mutual information across the three adjacent boundaries via a per-boundary loss decomposition

$$\mathcal{L}_{\text{C2HGI}} = \alpha_{c2p}\,\mathcal{L}_{c2p} + \alpha_{p2r}\,\mathcal{L}_{p2r} + \alpha_{r2c}\,\mathcal{L}_{r2c}, \quad (\alpha_{c2p}, \alpha_{p2r}, \alpha_{r2c}) = (0.4, 0.3, 0.3), \tag{1}$$

where each $\mathcal{L}_b$ is a bilinear-discriminator binary cross-entropy against random-shuffle negatives at boundary b.

Substrates are pre-trained per state on the full state-level check-in graph; the user-disjoint protocol applies to the downstream classifier and the per-fold transition prior, not to the substrate's exposure (§4.1). Two products feed our backbone after the encoder is frozen, both 64-dimensional: a per-check-in vector $\mathbf{e}_c$ from the bottom level for the category stream, and a per-region vector $\mathbf{e}_r$ from the third level for the region stream.

3.2 Cross-attention multi-task backbone

Feeding both tasks through one shared encoder forces an early compromise between two different feature distributions: per-check-in vectors drive next-category, while per-region vectors drive next-region. We instead keep two parallel input streams and let them exchange information through cross-attention before they meet a shared block. Per-task encoders (two-layer MLPs with LayerNorm and dropout) project the input streams $X^c, X^r \in \mathbb{R}^{L\times d}$ into $H^c, H^r \in \mathbb{R}^{L\times d_m}$ ($L = 9$, $d = 64$, $d_m = 256$). With $\text{MHA}(Q, K, V)$ scaled dot-product attention ($h = 8$ heads),

$$\widetilde{H}^c = H^c + \text{MHA}(H^c,\, H^r,\, H^r), \qquad \widetilde{H}^r = H^r + \text{MHA}(H^r,\, H^c,\, H^c), \tag{2}$$

so each stream queries the other. The cross-attended streams pass through a four-block shared residual backbone (LayerNorm, linear, LeakyReLU, dropout) before splitting into per-task heads (Figure 1).

The category head is a two-layer GRU over the cross-attended check-in stream with attention pooling and a seven-class softmax. The region head is an additive-prior STAN [8] variant: a STAN backbone over the cross-attended region stream, scored against learnable region embeddings, with an additive log-transition prior $\alpha \cdot \log T[r_{\text{last}}]$ (learnable α, built per fold from training-fold edges only). The log-prior pattern follows GETNext [15], but our head omits friendship and check-in graph priors and is not a faithful reproduction.

3.3 Training protocol

We aggregate the two task losses with a static weighting,

$$\mathcal{L} = \lambda_{\text{cat}}\,\mathcal{L}_{\text{cat}} + \lambda_{\text{reg}}\,\mathcal{L}_{\text{reg}}, \qquad \lambda_{\text{cat}} = 0.75, \quad \lambda_{\text{reg}} = 0.25, \tag{3}$$

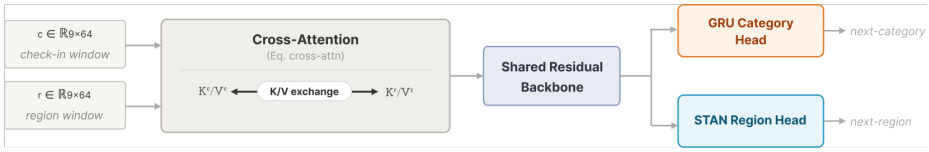

Fig. 1. Cross-attention multi-task backbone. Two task-specific input streams (X^c, X^r) feed a symmetric cross-attention block (Eq. 2); the bold black bidirectional K/V arrow is the cross-feed responsible for the encoder co-adaptation in Section 6.3, where a $\lambda = 0$ ablation does not isolate the targeted encoder. A shared residual backbone then splits into the GRU category head and the STAN-based region head with additive log-transition prior.

where $\mathcal{L}_{\text{cat}}, \mathcal{L}_{\text{reg}}$ are unweighted cross-entropy, fixed across all MTL rows. The drop-in adaptive balancers in §6.2 (FAMO [7], Aligned-MTL [11]) do not improve over this static choice, supporting the simpler weighting; a λ_{cat} sensitivity sweep is in the anonymous code release.

We optimise the model with AdamW over three parameter groups (category, region, shared). The learning-rate schedule depends on state scale: OneCycleLR with per-task-group alternating updates at FL/CA/TX, and a per-head constant learning rate at AL/AZ. The alternating step rescues the region head from saturation at large transition graphs, but costs category-side signal at smaller scales. We use a batch size of 2048 (1024 at the largest state), gradient accumulation of 2, and 50 epochs in FP32; per-task best checkpoints are selected on validation Acc@10 and macro-F1 for epochs ≥ 5. Exact per-state settings are in the anonymous code release.

Folds, seeds, and multi-seed pooling follow the user-disjoint StratifiedGroupKFold setup of Section 4; the same splits are reused across substrates, heads, and recipes so paired tests are well defined. Under cross-attention, $\lambda = 0$ ablation does not isolate the targeted encoder (it co-adapts through K/V projections); we use encoder-frozen isolation throughout (Section 6.3).

4 Experimental Setup

4.1 Datasets and preprocessing

We evaluate on five U.S.-state splits of the public *Gowalla* check-in dataset [4]: AL, AZ, FL, CA, TX. Each split is filtered to users with at least five check-ins, applied per state before fold creation so each state preserves its native cardinality. Per-state statistics are in Table 1; FL/CA/TX serve as the headline scale and AL/AZ as smaller-scale anchors. Region labels come from the per-state region partition used by the substrate engine.

For each user we form non-overlapping length-9 windows over the chronological sequence and predict the category and region of the tenth check-in (shorter sequences padded). Folds are **user-disjoint** via `StratifiedGroupKFold` with five folds (random_state=42), so train/val/test never share a user; splits are

Table 1. Dataset statistics for the five Gowalla [4] U.S.-state splits, after filtering to users with at least five check-ins. The headline three (FL/CA/TX) sit above the rule; the smaller-scale anchors (AL/AZ) sit below. Region cardinality is the prediction target on the next-region task.

State	Users	Check-ins	POIs	Regions	Mean traj. length
Florida (FL)	13,935	1,392,262	76,266	4,703	99.9
California (CA)	26,171	3,148,594	168,771	8,501	120.3
Texas (TX)	27,603	4,067,543	160,575	6,553	147.4
Alabama (AL)	1,622	109,695	11,666	1,109	67.6
Arizona (AZ)	3,331	227,956	20,469	1,547	68.4

reused across substrates, heads, and recipes so paired tests are well defined. The next-region transition prior is rebuilt per fold from training-fold edges only and seed-tagged. Check2HGI and HGI are pre-trained on each state's full check-in graph; the user-disjoint protocol applies to the downstream classifier and the per-fold transition prior but not to the substrate's exposure, so the setup is transductive at the substrate level.

4.2 Baselines and metrics

The headline contrast is the cross-attention multi-task model against matched-head single-task (STL) ceilings: a GRU category head and a STAN-based region head with an additive log-transition prior over training-fold-only edges [8]. **The same head classes appear on both sides of the contrast, so any difference between the multi-task and single-task rows is attributable to the sharing architecture, not to head capacity.** External baselines are POI-RGNN and MHA+PE (category) and Markov-1-region, single-task STAN, and ReHDM (region); we report them in §5.

Category metrics: macro-F1 (primary) and Acc@1/Acc@3 (top-k accuracy, secondary, since the 7-way distribution is skewed). Region metrics: Acc@10 (primary), Acc@5 and MRR (mean reciprocal rank, secondary). The joint score Δ_m [9,13] averages the per-task relative improvement of MTL over the matched-head STL ceiling (positive = MTL wins); we report it on a primary pair (macro-F1, MRR) and a secondary pair (macro-F1, Acc@10).

4.3 Statistical protocol

Paired contrasts use the paired Wilcoxon signed-rank test with `alternative='greater'` for directional claims and the two-one-sided test (**TOST**) for non-inferiority at $\delta \in \{2, 3\}$ pp. At the single-seed five-fold scale the max significance for $n = 5$ is $p = 0.0312$ (all folds agreeing).

Two ceiling protocols sit side by side. The **substrate axis** (Table 2) uses the seed-42 five-fold paired Wilcoxon at the $n = 5$ ceiling; multi-seed mean and

seed-σ are shown for Check2HGI at AL/AZ/FL but the paired test stays single-seed. The **architectural axis** (Tables 3, 4) pools four extra seeds $\{0, 1, 7, 100\}$ on top of seed-42, yielding $n = 20$ fold-pairs per state; FL's Δ_m pair additionally carries $n = 25$ fold-pairs. Effect sizes are in pp for F1, Acc@k, and MRR; Δ_m is in percent.

5 Results

We report results across five Gowalla state splits, organised around two findings: the substrate's effect is asymmetric across the two tasks (§5.1), and joint multi-task training over Check2HGI adds a small category lift while paying the expected region cost (§5.2). Section 5.3 reports the joint Δ_m score; Section 5.4 situates the model against external baselines.

5.1 Substrate ablation: the embedding is asymmetric across tasks

On next-category, per-visit context lifts macro-F1 by +14.18 to +29.02 percentage points (pp) at every state (smallest at AZ, largest at FL). Table 2(a) reports the numbers under the matched-head GRU single-task ceiling, with substrate as the only changing factor. The paired Wilcoxon signed-rank test reaches the $n = 5$ ceiling everywhere ($p = 0.0312$, 5/5 folds positive). Check2HGI macro-F1 is stable across four extra seeds $\{0, 1, 7, 100\}$ at AL/AZ/FL with seed-$\sigma \leq 0.17$ pp. Two robustness probes back this: the AL/AZ gap is head-invariant across linear, single-layer, GRU, and LSTM probes (in the anonymous code release), and the FL/CA/TX gap replicates under the matched-head ceiling. We read the lift as evidence that per-visit context, not POI-level attribute encoding, is what the next-category head exploits.

On next-region, by contrast, per-place HGI ties or marginally exceeds Check2HGI. Table 2(b) reports Acc@10 under the matched-head STAN-based single-task ceiling. HGI is nominally above Check2HGI by 1.6 to 3.1 pp at every state; on FL/CA/TX the gap shrinks to 1.6 to 2.1 pp and TOST non-inferiority at $\delta = 2$ pp passes at CA and TX. At FL the $\delta = 3$ pp test passes. AL and AZ stay outside both margins. The short-horizon transition structure driving the region head is captured equally well, or fractionally better, by per-place embeddings whose category-level pooling smooths the per-visit variance that helps the category head.

The substrate's effect is asymmetric across the two tasks. The category head reads per-visit context and lifts by an order of magnitude more than its standard-deviation envelope; the region head reads transition structure that is preserved under per-place pooling and ties within ± 2 pp. Section 6 attributes the asymmetry through a per-visit-context counterfactual replicated at AL and AZ. Any multi-task system on this task pair therefore operates on a substrate that already encodes the easier task more clearly than the harder one.

Table 2. Substrate ablation under matched-head single-task ceilings (5-fold CV, seed-42). Cells are mean ± fold-σ; Δ = Check2HGI − HGI; paired Wilcoxon reaches $p = 0.0312$ when 5/5 folds agree. (a) Next-category macro-F1 with a GRU head; asterisk marks 5/5 folds positive. (b) Next-region Acc@10 with a STAN-based head + additive log-transition prior; TOST verdicts at $\delta \in \{2, 3\}$ pp test non-inferiority (✓) or not (×).

(a) Next-category macro-F1.

State	HGI cat F1	Check2HGI cat F1	Δ (pp)	Wilcoxon p
Alabama (AL)	25.26 ± 1.18	40.76 ± 1.50	+15.50*	0.0312
Arizona (AZ)	28.99 ± 0.51	43.17 ± 0.28	+14.18*	0.0312
Florida (FL)	34.41 ± 1.05	63.43 ± 0.98	+29.02*	0.0312
California (CA)	31.13 ± 1.04	59.94 ± 0.59	+28.81*	0.0312
Texas (TX)	31.89 ± 0.55	60.24 ± 1.84	+28.34*	0.0312

(b) Next-region Acc@10.

State	HGI Acc@10	Check2HGI Acc@10	Δ (pp)	TOST $\delta = 2$	TOST $\delta = 3$
Alabama (AL)	61.86 ± 3.29	59.15 ± 3.48	−2.71	×	×
Arizona (AZ)	53.37 ± 2.55	50.24 ± 2.51	−3.13	×	×
Florida (FL)	71.34 ± 0.64	69.22 ± 0.52	−2.12	×	✓
California (CA)	57.77 ± 1.12	55.92 ± 1.20	−1.85	✓	✓
Texas (TX)	60.47 ± 1.26	58.89 ± 1.28	−1.59	✓	✓

5.2 Multi-task vs single-task on the per-visit substrate

Joint multi-task training is positive on next-category at four of five states. Table 3 reports the main contrast: the cross-attention multi-task backbone against the matched-head GRU and STAN-based single-task ceilings on Check2HGI, multi-seed pooled at $n = 20$ fold-pairs per state. Category is positive at FL +1.40 pp ($p = 2\text{e}{-6}$), CA +1.68 pp ($p = 2\text{e}{-6}$), TX +1.89 pp ($p = 2\text{e}{-6}$) and AZ +1.20 pp ($p < 1\text{e}{-4}$). At AL the contrast is statistically significant but small in magnitude ($\Delta = -0.78$ pp, $p = 0.036$, $n = 20$, under 2 % relative on the 41 % F1 scale). The four positive states sit outside the seed and fold envelope; we report the AL negative as is rather than absorbing it.

Next-region follows the cost-versus-gain pattern reported in the dense-prediction MTL line [13]. MTL pays −7.34 pp (FL) to −16.59 pp (TX) on Acc@10 at every state, all statistically significant ($p \leq 1.9\text{e}{-6}$). The pattern is broadly downward on AL → AZ → FL, with CA preserving the FL regime and TX breaking the monotonic shrink, plausibly under state-specific factors beyond raw class count (geographic POI distribution, user-density skew, per-state log-transition prior coverage); we do not read the AL → FL contraction as an inferential scaling claim.

On Acc@10, the multi-task backbone trades the matched-head STAN-based ceiling for a regime that sits below a one-step Markov-1-region floor at four of five states (AL is the only state above the floor; per-state pp gaps: AL +3.16, AZ −2.18, FL −1.78, CA −4.74, TX −12.10). The matched-head Check2HGI

Table 3. Cross-attention multi-task model vs. matched-head single-task ceilings on Check2HGI. Multi-seed pooled at $n = 20$ fold-pairs per state (seeds $\{0, 1, 7, 100\} + 42$, five user-disjoint folds). Cells are mean ± pooled σ; Δ = MTL − STL in pp; paired one-sided Wilcoxon p-values. States ordered by scale (AL smallest, TX largest).

	Next-category (macro-F1)				Next-region (Acc@10)			
State	STL	MTL	Δ	p	STL	MTL	Δ	p
AL	41.35±0.17	40.57±0.24	−0.78	0.036	61.21±0.18	50.17±0.24	−11.04	1.9e−6
AZ	43.90±0.17	45.10±0.19	+1.20	<1e−4	53.06±0.15	40.78±0.07	−12.27	1.9e−6
FL	67.16±0.13	68.56±0.79	+1.40	2e−6	70.62±0.09	63.27±0.10	−7.34	1.9e−6
CA	62.39±0.13	64.07±0.14	+1.68	2e−6	56.85±0.09	47.35±0.11	−9.50	1.9e−6
TX	63.11±0.13	65.00±0.11	+1.89	2e−6	59.44±0.09	42.84±0.14	−16.59	1.9e−6

Δ in percentage points (multi-task − single-task).

ceiling sits +5 to +14 pp above Markov-1 at every state, so the cross-attention region head trades that ceiling for the joint regime, underperforming on transition structure a one-step Markov chain already captures. The Δ_m MRR axis at FL (+2.33 %, $n = 25$, statistically significant in §5.3) characterises the joint tradeoff more cleanly than Acc@10 in this regime; we report both.

With the substrate fixed at the per-visit granularity, the multi-task backbone delivers a small category lift at four of five states and pays a sign-consistent cost on the harder head at every state, the classic multi-task tradeoff. The cost is structural to the cross-attention shared-backbone interaction with the region head; Section 6 reports drop-in replacements (FAMO, Aligned-MTL, hierarchical-additive-softmax region head) that fail to close the next-region gap.

5.3 Joint Δ_m: broadly negative; FL on MRR the exception

On the joint Δ_m score [9,13] (the multi-task model's percent deviation from the matched-head ceilings averaged across tasks), FL is the only Pareto-positive cell on the primary axis at multi-seed; the other four states are negative on both axes. On the primary pair (category macro-F1, region MRR), FL carries an $n = 25$ multi-seed sweep and reaches Δ_m-MRR = +2.33 % at $p = 2.98\text{e}{-8}$, 25/25 fold-pairs positive. The secondary pair at FL is Δ_m-Acc@10 = −1.12 % at $p = 3.20\text{e}{-5}$ (4/25); MRR rewards near-correct candidates that raw top-10 accuracy misses, consistent with the Acc@10-saturated regime above. The other four states report at the $n = 5$ Wilcoxon ceiling, with Δ_m negative on both axes (MRR from −24.84 % to −1.61 %; Acc@10 from −22.41 % to −6.85 %), at most 1/5 fold-pairs positive.

5.4 External baselines

Against external baselines, the matched-head STL category ceiling on Check2HGI roughly doubles the strongest user-disjoint POI-RGNN reproduction, while the multi-task region row trails the strongest external region row. Table 4 reports the comparison.

Table 4. External-baseline comparison at FL/CA/TX. Italicised rows are matched-head STL ceilings on Check2HGI; the proposed row is our MTL model. POI-RGNN and MHA+PE are user-disjoint reproductions; STAN is the STAN backbone [8] on our HGI substrate with our training-fold-only log-prior. STAN/ReHDM run at FL only (ReHDM scales quadratically in region cardinality).

	Category macro-F1			Region Acc@10		
Method	FL	CA	TX	FL	CA	TX
External baselines (faithful reproductions)						
Markov-1 (floor)	23.98±1.22	19.97±1.77	18.17±1.38	65.05±0.93	52.09±0.80	54.94±0.46
POI-RGNN [1]	34.49±1.14	31.78±0.82	33.03±0.70	–	–	–
MHA+PE [16]	32.06±0.23	29.13±0.71	29.91±0.43	–	–	–
STAN [8] (HGI)	–	–	–	**73.58±0.43**	**60.45±0.97**	**62.70±0.37**
ReHDM [6]	–	–	–	65.68±0.26	–†	–†
Matched-head single-task ceilings (Check2HGI substrate)						
STL category ceiling	67.16±0.13	62.39±0.13	63.11±0.13	–	–	–
STL region ceiling	–	–	–	70.62±0.09	56.85±0.09	59.44±0.09
Proposed cross-attention multi-task model						
MTL (ours)	**68.56±0.79**	**64.07±0.14**	**65.00±0.11**	63.27±0.10	47.35±0.11	42.84±0.14

Bold cells: best per column. † ReHDM not reported at CA/TX (compute budget).

On next-category, our matched-head STL ceiling on Check2HGI (67.16 at FL) sits well above our user-disjoint POI-RGNN reproduction [1] (34.49); the published POI-RGNN evaluation used non-user-disjoint folds, so absolute numbers are not directly comparable. §6.1 attributes ∼72% / ∼64% of the AL/AZ substrate gap specifically to per-visit context, so most of the lift is on the substrate axis, not from multi-task aggregation. On next-region, the matched-head STAN-based head with additive log-prior on the HGI substrate is the strongest external row at this scale; per-place HGI is a more natural fit for the region head, consistent with the substrate-axis reading. Our multi-task row trails this row by −10 to −20 pp on Acc@10, the substrate gap compounded by the cross-attention region cost.

6 Mechanism and Ablations

This section attributes the substrate gain through a per-visit counterfactual, stress-tests the next-region cost against drop-in alternatives, and closes with a methodological note on cross-attention loss-side ablations.

6.1 Per-visit context counterfactual (five states)

To isolate per-visit context, we compare canonical Check2HGI against POI-pooled Check2HGI (canonical vectors mean-pooled per POI and applied uniformly to every visit, killing per-visit variation while preserving the training

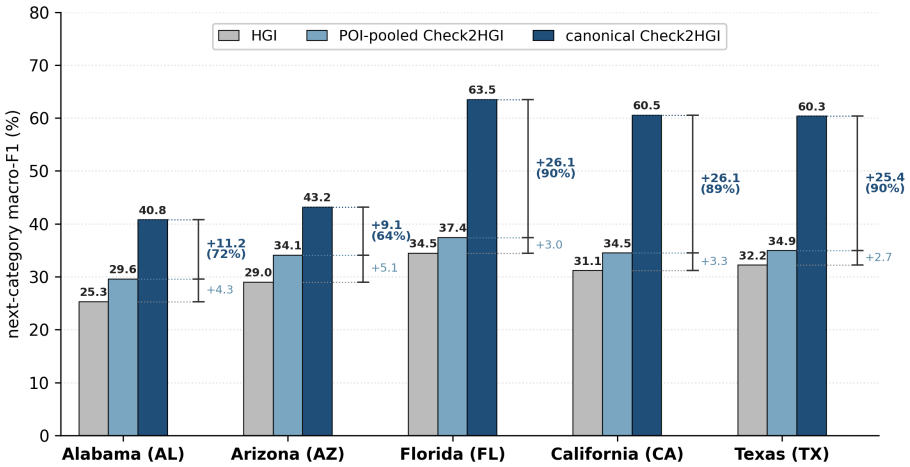

Fig. 2. Per-visit context decomposition at five states under the matched-head GRU STL ceiling. The substrate gap (canonical Check2HGI − HGI) splits into a per-visit component (canonical − POI-pooled) and a training-signal component (POI-pooled − HGI). Per-state shares are reported in §6.1.

signal). Canonical vs. pooled isolates per-visit context; pooled vs. HGI isolates the contrastive training signal.

Under the matched-head GRU single-task ceiling, the substrate gap (canonical − HGI) splits consistently at all five states (Figure 2): per-visit context dominates at every state, with both components positive. Paired Wilcoxon signed-rank tests reach the $n = 5$ ceiling at AL and AZ ($p = 0.0312$, 5/5 folds positive); FL/CA/TX show the same monotone ordering at all five per-fold pairs. A two-band pattern emerges across state scale: at small states (AL ∼72 %, AZ ∼64 %) the training-signal residual is 28–36 %; at large states (FL/CA/TX, ∼89–90 %) the pooled embeddings collapse almost entirely to the HGI level, leaving ∼10 % to the training signal. This is consistent with large transition graphs (4.7k–8.5k regions) providing weaker per-POI contrastive signal, so Check2HGI's advantage at large states is almost entirely in the per-visit contextual variation it introduces.

6.2 Drop-in MTL ablations do not recover the next-region cost

The next-region cost is negative at every state (7 to 17 pp on Acc@10). At FL we tried three drop-in replacements (FAMO [7], a magnitude balancer; Aligned-MTL [11], a direction-aligning balancer; and a hierarchical-additive-softmax region head) with substrate, backbone, optimiser and schedule fixed at §3. None reached the +3 pp Acc@10 recovery target or paired-Wilcoxon $p_{\text{greater}} < 0.10$; the strongest, the hierarchical head, reaches only $\approx +1.7$ pp. The cost stays

intact, suggesting a structural cost of the cross-attention interaction with the region head rather than a loss-recipe artefact.

6.3 Cross-attention $\lambda = 0$ is not encoder isolation

A natural ablation zeros one task's weight in Eq. (3), on the assumption that $\lambda_{\text{cat}} = 0$ or $\lambda_{\text{reg}} = 0$ ablates the corresponding encoder. Under cross-attention this is unsound: the targeted encoder still updates through the key/value projections of the surviving stream, so its parameters co-adapt to the surviving task. A targeted probe at AZ quantifies the gap: encoder-frozen isolation reaches region Acc@10 60.72, while loss-side $\lambda_{\text{cat}} = 0$ reaches 62.70 on the same folds, a +1.98 pp lift driven through the K/V projections. We therefore use **encoder-frozen isolation** (targeted encoder random-init, frozen) throughout; the caution generalises to any cross-task attention multi-task model.

7 Discussion and Limitations

Overall, the multi-task picture is dominated by the substrate's asymmetric value across the two tasks, not by cross-task transfer through joint training. Per-visit context is statistically significant for next-category at every state (+14 to +29 pp matched-head); on next-region the same substrate ties or marginally trails per-place HGI (1.6 to 3.1 pp; TOST $\delta = 2$ pp passes at CA and TX). With the substrate fixed, joint training adds a small category lift at four of five states (the AL contrast is statistically significant but under 2 % relative) and pays a 7 to 17 pp cost on next-region Acc@10 at every state. The conventional reading where joint training transfers signal to lift both heads does not hold on the harder task: the category lift survives the substrate ablation, and the region cost survives the substrate fix. The substrate's value is also large in absolute terms (matched-head STL on Check2HGI reaches 67.16 at FL, roughly doubling the strongest user-disjoint POI-RGNN reproduction at 34.49) and is recoverable by any downstream architecture that consumes per-visit context. The substrate is the dominant factor; multi-task coupling adds a small category gain at the cost of region accuracy.

Four limitations apply. The per-visit decomposition replicates at AL+AZ but is not generalised to the remaining three states, and the drop-in panel ran at FL only. On Acc@10 the multi-task backbone trails the Markov-1-region floor at four of five states (§5.2); we read this as a structural cost of joint training on the harder head, and report Δ_m MRR alongside (FL +2.33 %, $n = 25$, highly significant). External coverage is asymmetric: ReHDM [6] is reported at FL only and a literature-faithful STAN reproduction is omitted (Table 4 caption). The substrate is transductive at the state level (§4); held-out-user substrate retraining, cross-task transfer (e.g. next-POI), and architectural extensions for the region head are left to follow-up work.

8 Conclusion

We introduced Check2HGI, a check-in-level contextual graph embedding, and tested whether it carries the multi-task POI prediction win across five U.S.-state Gowalla splits. The substrate is asymmetric across the two tasks: per-visit context lifts matched-head next-category macro-F1 by +14 to +29 pp at every state, with the AL/AZ counterfactual attributing ∼72 % and ∼64 % of the gap to per-visit context, while per-place embeddings tie or marginally exceed Check2HGI on next-region. With the substrate fixed, joint multi-task training adds a small category lift at four of five states and pays a 7 to 17 pp region cost at every state; drop-in balancers and a head-side fix do not close the gap. Three open lines follow: asymmetric routing across the shared backbone to relax cross-attention interference, encoder enrichment fusing per-visit contrast with the explicit spatial-temporal sub-encoders of [10], and a region-side dynamic prior conditioned on the user's recent trajectory.

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